Latest version: vuphan314.github.io
This version: 2020/10/14

Vu Hoang Nguyen Phan

Curriculum Vitae

vuphan314

Vu Phan is a Computer Science Ph.D. student of Prof. Moshe Vardi at Rice University in Texas. He applies knowledge compilation and dynamic programming to counting problems. He expects to graduate in 2022 Spring.

Education

2017–2022 **Doctor of Philosophy**, *Computer Science*, Rice University, GPA 3.57.

2017–2019 **Master of Science**, *Computer Science*, Rice University, GPA 3.60.

2014–2017 **Dual Bachelor of Science**, *Computer Science & Math*, Texas Tech University, GPA 4.00.

Bibliography

Vu Phan is the corresponding author of the following publications (authors sorted by surnames), except for [RP18].

Thesis

[Pha19] Vu H. N. Phan. "Weighted Model Counting with Algebraic Decision Diagrams". MS thesis. Rice University, 2019. URL: scholarship.rice.edu/handle/1911/107761.

Conference Papers

[DPV20a] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "DPMC: Weighted Model Counting by Dynamic Programming on Project-Join Trees". In: *International Conference on Principles and Practice of Constraint Programming (CP)*. 2020. URL: arxiv.org/abs/2008.08748.

[DPV20b] Jeffrey M. Dudek, Vu H. N. Phan, and Moshe Y. Vardi. "ADDMC: Weighted Model Counting with Algebraic Decision Diagrams". In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2020. URL: arxiv.org/abs/1907.05000.

[RP18] Nelson Rushton and Vu H. N. Phan. "A Discrete Math Methodology for Computer Science". In: *International Conference on Frontiers in Education: Computer Science and Computer Engineering (FECS)*. 2018. URL: csce.ucmss.com/cr/books/2018/LFS/CSREA2018/FEC4172.pdf.

Workshop Paper

[Pha18] Vu H. N. Phan. "Syntactic Conditions for Antichain Property in Consistency Restoring Prolog". In: *Workshop on Answer Set Programming and Other Computing Paradigms (ASPOCP)*. 2018. URL: arxiv.org/abs/1809.09319.

Professional Service

2020 **Peer Reviewer**, *International Conference on Artificial Intelligence and Statistics (AISTATS)*.

Software Development

2019–2020 **DPMC**, *Dynamic Programming for Model Counting*, `github.com/vardigroup/DPMC`.
- Implemented a competitive weighted #SAT solver with Jeffrey Dudek in [DPV20a].

Supervisor: Moshe Vardi (Rice University).

2018–2020 **ADDMC**, *Algebraic Decision Diagrams for Model Counting*, `github.com/vardigroup/ADDMC`.
- Implemented an exact weighted #SAT solver for conjunctive normal form in [DPV20b; Pha19].

Supervisor: Moshe Vardi (Rice University).

2016–2017 **LED**, *Literate Programming*, `vuphan314.github.io/LED`.
- Translated the programming language `LED` in [RP18] into `SequenceL` and `LaTeX`.

Supervisor: Nelson Rushton (Texas Tech University).

2015–2016 **L**, *Logic Programming*, `github.com/iensen/LtoASPtranslator`.
- Translated the progrmaming language `L` into `Answer Set Prolog` with Evgenii Balai.

Supervisor: Nelson Rushton (Texas Tech University).

Programming Languages

Procedural **Python, C++, Java**.

Functional **Scala, Scheme**.

Graduate Courses

2020 **Fall semester**.
- Bioinformatics: Sequence Analysis

2020 **Spring semester**.
- Automated Program Verification (topic: Reinforcement Learning)
- Statistical Machine Learning

2019 **Fall semester**.
- Computer Systems Architecture

2019 **Spring semester**.
- Automated Program Verification (topic: Autonomous Driving)
- Compiler Construction
- Multi-Core Computing

2018 **Fall semester**.
- Artificial Intelligence

2018 **Spring semester**.
- Automated Program Verification (topic: Reactive Synthesis)
- Principles of Programming Languages

2017 **Fall semester**.
- Advanced Logic in Computer Science

Scholarships

2017–2022 **Ph.D. Student Tuition & Stipend**, *Computer Science Department*, Rice University.

2015–2017 **Academic Achievement Scholarship**, *College of Arts & Sciences*, Texas Tech University.

2014–2016 **Proven Achiever Scholarship**, *Financial Aid Office*, Texas Tech University.